

2.)

$$G(s) \cdot X(s) = \frac{1}{s^2 + 3s + 2}$$

$$x(t) = e^{-3t} \sigma(t) \rightarrow \text{Laplace}$$

$$\frac{1}{s+3}$$

$$x(t) = e^{-3t} \sigma(t) \rightarrow X(s) = \text{LT} \{e^{-3t} \sigma(t)\}$$

$$X(s) = \int_{-\infty}^{\infty} e^{-3t} \sigma(t) e^{-st} dt = \int_0^{\infty} e^{-3t} e^{-st} dt$$

$$= \int_0^{\infty} e^{+(-3-s)t} dt = \left(\frac{1}{-3-s} \right) \cdot e^{+(-3-s)t} \Big|_0^{\infty} = \frac{1}{-3-s} (e^{-\infty} - e^0) = \frac{1}{3+s}$$

$$X(s) \cdot G(s) = \frac{1}{3+s} \cdot \frac{2}{s^2+3s+2} = \frac{2}{(s+3)(s^2+3s+2)}$$

$$s^2+3s+2=0$$

$$s_{1,2} = -\frac{3}{2} \pm \sqrt{\frac{9}{4} - 2} = -\frac{3}{2} \pm \sqrt{\frac{1}{4}} = -\frac{3}{2} \pm \frac{1}{2}$$

$$s_1 = -1$$

$$s_2 = -2$$

$$\frac{2}{(s+3)(s+1)(s+2)} = \frac{R_1}{s+1} + \frac{R_2}{s+2} + \frac{R_3}{s+3} \quad | \cdot (s+3)$$

$$\frac{2 \cdot (s+3)}{(s+3)(s+1)(s+2)} = R_3 + \left(\frac{R_1}{s+1} + \frac{R_2}{s+2} \right) \cdot (s+3)$$

$$\frac{2}{(-1)(-2)} = R_3 = \frac{2}{(-2)(-1)} = 1$$

$$\frac{2}{(s+3)(s+2)} = R_1 + \left(\frac{R_2}{s+2} + \frac{R_3}{s+3} \right) (s+1)$$

$$\frac{2}{2 \cdot 1} = R_1 = 1$$

$$\frac{2}{(s+1)(s+3)} = R_2 + \left(\frac{R_1}{s+1} + \frac{R_3}{s+3} \right) (s+2)$$

$$\frac{2}{1 \cdot (-1)} = R_2 = -2 \Rightarrow \frac{1}{s+1} - \frac{2}{s+2} + \frac{1}{s+3}$$

$$Y(s) = \frac{1}{s+1} - \frac{2}{s+2} + \frac{1}{s+3}$$

$$y(t) = \text{LT}^{-1} \{Y(s)\} = (e^{-t} - 2e^{-2t} + e^{-3t}) \sigma(t)$$

Bsp 3)

$$\int_{-\infty}^{\infty} x^2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

$$x(t) = r(t)$$

$$r(t) \rightarrow T \cdot \text{si}(\omega \frac{T}{2})$$

B... Bandbreite
Höhe... 1

$$\int_{-\infty}^{\infty} r^2(t) dt = \int_{-\frac{T}{2}}^{\frac{T}{2}} 1^2 dt = \left. t \right|_{-\frac{T}{2}}^{\frac{T}{2}} = \frac{T}{2} - (-\frac{T}{2}) = \underline{T}$$

Fläche des Rechtecks ist T

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \underbrace{T^2}_{\text{Fläche wird ersetzt}} \cdot \text{si}^2(\omega \frac{T}{2}) d\omega$$

Breite ist $2\omega_B$

$$= \frac{1}{2\pi} \cdot \underbrace{T^2 \cdot \omega_B \cdot 2}_{\text{Rechteckfläche}} = \frac{1}{\pi} \cdot T^2 \cdot \omega_B$$

Flächen gleichsetzen

$$T = \frac{1}{\pi} \cdot T^2 \cdot \omega_B$$

$$\omega_B = \frac{\pi}{T}$$

$$\underline{B} = \frac{\omega_B}{2\pi} = \frac{\frac{\pi}{T}}{2\pi} = \underline{\underline{\frac{1}{2T}}}$$

Bsp 4)

$$\dot{x}(t) = \overbrace{Ax(t)} + \overbrace{b u(t)}$$

$$\dot{x}(t) = \begin{bmatrix} -2 & 1 \\ 0 & -1 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \underbrace{\begin{bmatrix} 1 & 1 \end{bmatrix}}_{C^T} x(t)$$

$$G(s) = \frac{Y(s)}{U(s)} \rightarrow \begin{matrix} \text{Ausgangssignal} \\ \text{Eingangssignal} \end{matrix}$$

Die Übertragungsfunktion

$$\dot{x}(t) = Ax(t) + b u(t)$$

$$sX(s) - x(0) = AX(s) + bU(s)$$

$$X(s)(sE - A) = bU(s) + x(0)$$

$$X(s) = (sE - A)^{-1} b U(s) + (sE - A)^{-1} x(0)$$

$$\begin{aligned} \text{NR: } (sE - A)^{-1} &= \left(\begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} -2 & 1 \\ 0 & -1 \end{bmatrix} \right)^{-1} = \begin{pmatrix} s+2 & -1 \\ 0 & s+1 \end{pmatrix}^{-1} \\ &= \frac{1}{|sE - A|} \begin{bmatrix} s+1 & +1 \\ 0 & s+2 \end{bmatrix} \Rightarrow Z(s) \end{aligned}$$

$$X(s) = \underbrace{\frac{1}{|sE - A|} \cdot Z(s) \cdot b}_{G(s)} U(s) + \frac{1}{|sE - A|} \cdot Z(s) \cdot x(0)$$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{1}{|sE - A|} \cdot Z(s) \cdot b = \frac{1}{|sE - A|} \cdot \begin{bmatrix} s+1 & 1 \\ -1 & s+2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1 + s + 2}{|sE - A|}$$

$$= \frac{s+3}{|sE - A|}$$

$$|sE - A| = \begin{vmatrix} s+2 & -1 \\ 0 & s+1 \end{vmatrix} =$$

$$(s+2) \cdot (s+1) - 0 = (s^2 + 2s + s + 2) = s^2 + 3s + 2$$

$$= \frac{s+3}{s^2 + 3s + 2}$$

Die Transitionsmatrix

$$e^{At} \rightarrow (sE - A)^{-1}$$

aufpassen auf Faktoren!

$$x(t) = e^{At} b u(t) + e^{At} \cdot x(0)$$

$$y(t) = x(t) \cdot c^T = c^T (e^{At} b u(t) + e^{At} \cdot x(0)) = c^T$$

$$y(t) = [1 \ 1] \cdot e^{At} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \cdot \delta(t) + e^{At} \cdot x(0) \cdot [1 \ 1]$$

$$G(w) = \frac{s+3}{s^2+3s+2}$$

$$U(w) = 1$$

$$s^2+3s+2=0$$

$$s_{1,2} = \frac{-3 \pm \sqrt{9-8}}{2} \Rightarrow \begin{matrix} s_1 = -1 \\ s_2 = -2 \end{matrix}$$

$$\frac{s+3}{(s+1)(s+2)} = \frac{R_1}{s+1} + \frac{R_2}{s+2} \quad | \cdot (s+2)$$

$$s=-2: \frac{s+3}{s+1} = R_2 \rightarrow \frac{1}{-1} = R_2 = -1$$

$$s=-1: \frac{s+3}{s+2} = R_1 \rightarrow \frac{2}{1} = R_1 = 2$$

$$Y(s) = \frac{2}{s+1} - \frac{1}{s+2} \Rightarrow$$

$$y(t) = 2e^{-t} - e^{-2t}$$

$$\begin{bmatrix} -2 & 1 \\ 0 & -1 \end{bmatrix}$$

?

$$\begin{aligned} y(t) &= c^T \cdot e^{At} \cdot b \\ &= [1 \ 1] \cdot e^{At} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ &= [1 \ 1] \cdot e^{\begin{bmatrix} -2 & 1 \\ 0 & -1 \end{bmatrix} t} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = [1 \ 1] \cdot \begin{bmatrix} e^{-t} \\ e^{-t} \end{bmatrix} \\ &= \underline{e^{-t} + e^{-t}} \end{aligned}$$

$$(sE - A)^{-1} = \frac{1}{(s+1)(s+2)} \begin{bmatrix} s+1 & 1 \\ 0 & s+2 \end{bmatrix} = \begin{bmatrix} \frac{1}{s+2} & \frac{1}{(s+1)(s+2)} \\ 0 & \frac{1}{s+1} \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} e^{-2t} & e^{-t} - e^{-2t} \\ 0 & e^{-t} \end{bmatrix}$$

$$y(t) = c^T \cdot e^{At} \cdot b = [1 \ 1] \cdot \begin{bmatrix} e^{-2t} & e^{-t} - e^{-2t} \\ 0 & e^{-t} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= [1 \ 1] \begin{bmatrix} e^{-2t} - e^{-t} \\ e^{-t} \end{bmatrix} = e^{-2t} - e^{-t} + e^{-t}$$

$$\underline{y(t) = 2e^{-t} - e^{-2t}}$$